GLR-CNN: CNN-Based Framework With Global Latent Relationship Embedding for High-Resolution Remote Sensing Image Scene Classification

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***Abstract*— High-resolution remote sensing image (HRSI) scene classification often faces challenges; for example, the intraclass similarity is low, but the interclass similarity is high due to complex backgrounds and variable scene scales. Convolutional neural networks (CNNs), the leading methods for HRSI scene classification, offer excellent performance. However, traditional CNNs require fixed-size inputs, which are a limitation when dealing with HRSI that represent large image domains, potentially degrading classification performance. To overcome these problems, we propose a CNN-based model named GLR-CNN in this article. First, to capitalize on the information from large-scale scenes adequately, VGG16 is utilized to extract the deep representative features, fine-tuned by the target HRSI of any size. Furthermore, a multilayer feature fusion block based on the channel–spatial attention algorithm is integrated into the CNN to capture more discriminative features from arbitrary-size images. Finally, to enhance the consistency between image features and similarities, a global latent relationship is used to measure the similarities among image features, then embed it into the fully connected layers (FCLs), and construct the latent relationship constraint. The model is optimized by the joint objective function including the latent relationship constraint and cross-entropy loss with label smoothing. Extensive experiments on three HRSI datasets obtained improvements of 3.93%, 6.5%, and 2.2% in overall accuracy compared to the fine-tuned VGG16 model, proving the effectiveness of the GLR-CNN method.**

***Index Terms*— Convolutional neural networks (CNNs), global latent relationship, joint objective function, multilayer feature fusion block, remote sensing image scene classification.**

NOMENCLATURE

λ	It is used to balance the importance of the latent relationship constraint.
C	Number of channels in the target features.
L	Number of FCLs.
N	Number of samples in a batch.
n	Number of samples.
O, P	Size of target features.
σ, ReLU	Nonlinear function.
b	Bias matrix of the network.
h	Feature matrix of the network.
$\mathbf{L}^*$	Laplacian matrix.
R	Latent relationship matrix.
W	Weight matrix of the network.
X	Database images.
tr	Trace of matrix.

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I. INTRODUCTION

RECENTLY, the easy availability of remote sensing images from multiple sensors has been a direct result of advancements in sensor technology. The geographical information with a high spatial resolution is captured from the high-resolution remote sensing images (HRSI), which are important for exploring the surface of the Earth, particularly in analyzing scene-level land use. Consequently, there is a pressing need for research into scene-level identification and classification of HRSI. HRSI scene classification (HRSI-SC) involves categorizing images into predefined semantic labels based on their content, significantly benefiting various applications, e.g., urban planning [1], [2], disaster monitoring [3], and geospatial object detection [4], [5].

However, HRSI-SC presents challenges due to low intraclass similarity and high interclass similarity (LH-CS), stemming from the variability in ground object scales and the presence of diverse ground objects. Recent efforts to improve HRSI-SC have been substantial; typically, classification methods are categorized into handcrafted feature extraction, unsupervised feature learning, and deep feature learning [6]. Handcrafted features, derived from human domain expertise and engineering skills, include texture descriptors [7], color histograms [8], scale-invariant feature transform [9], and histograms of oriented gradients [10]. The above methods, however, often suffer from human involvement, leading to inadequate representation of image features. To improve the scene classification performance, unsupervised

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feature-learning methods have been proposed, focusing on deriving features from unlabeled data to allow for more discriminative learning with limited datasets. Notable methods include principal component analysis (PCA) [11] and autoencoder models [12], which have shown effectiveness in HRSI-SC. However, with the evolution of deep learning models, especially convolutional neural networks (CNNs), the prominence of deep feature learning has grown. Compared with the unsupervised feature-learning method, CNNs are excellent in learning more representative and discriminative features, even from insufficient samples, prompting many CNN-based methods for HRSI-SC [13], [14], [15].

CNN-based feature-learning approaches have achieved excellent performance in image classification. For example, Chen et al. [16] combined the FCN model and attention mechanism to recognize the noncloud-and-cloud scenes and achieve the remote sensing image classification under noncloud areas and cloud areas, respectively, to improve the performance of urban green space mapping in cloud-land cover areas. In recent years, a variety of CNN-based models have been developed to address the LH-CS problem in HRSI-SC. Initially, these methods capitalize on the features of pretrained CNNs, subsequently enhancing the feature-learning capabilities of the model with novel techniques to reduce the influence of the LH-CS problem on HRSI-SC [17]; for example, Zheng et al. [18] combined the pretrained CNN model and multiscale pooling module to improve the performance of aerial scene classification. However, these approaches have limitations, i.e., pretrained CNNs require fixed-size inputs, necessitating the resizing of HRSI to fit these dimensions, which can result in the loss of crucial information given the large-scale nature of these images. Thus, how to extract the key and useful features from large-scale HRSI scenes and overcome the LH-CS problem effectively is an urgent problem to be solved in HRSI-SC.

To mitigate these problems, we introduce the GLR-CNN, a novel model based on a deep CNN with global latent relationship embedding. First, a deep pretrained CNN is used to extract deep features from the HRSI without resizing, to avoid introducing noise. Second, a multilayer feature fusion block (MFF-CNN) based on the channel–spatial attention algorithm is designed to extract and integrate key information from multiple levels of the model. Furthermore, the global latent relationship is embedded into the fully connected layers (FCLs) to mitigate the LH-CS problems by utilizing a manifold learning algorithm to enhance consistency between feature and latent relationship. The objective function comprises cross-entropy loss based on label smoothing and a latent relationship constraint. Experimental results on HRSI datasets show that GLR-CNN achieves superior accuracy compared to most CNN-based methods. The major contributions of this work are given as follows.

1) For extracting useful features from large-scale HRSI, a MFF-CNN block based on the channel–spatial attention algorithm is applied to HRSI-SC to robustly learn and integrate discriminative image features from different levels.
2) A global latent relationship constructed from the similarity based on the global distance distribution is embedded into the model to enhance the consistency between the feature and latent relationship to mitigate the impact of the LH-CS problem on HRSI-SC.
3) A novel feature-learning CNN-based model is proposed to overcome the shortcomings of previous CNNs by combining cross-entropy loss based on label smoothing with latent relationship constraints. The outstanding performance of the GLR-CNN model across the three HRSI datasets confirms that it is an effective model for HRSI-SC.

For clarity, we illustrate important notation and definitions used throughout this article in the Nomenclature.

II. Related Work

A. CNNs for HRSI Scene Classification

Deep learning, especially CNNs, has excelled in the field of HRSI-SC, with numerous excellent methods being proposed. A multistage self-guided separation network, based on CNNs, is proposed in [13], utilizing target background information to separate the target and background for overcoming the LH-CS problem. Cheng et al. [21] developed D-CNN models to improve baseline CNN performance, including AlexNet [19] and VGG16 [20], by incorporating metric learning regularization into a new discriminative objective function to overcome the LH-CS issue. Zhang et al. [22] proposed the CNN-CapsNet model that uses CNNs for hierarchical feature extraction and CapsNet for encoding feature properties and spatial information. He et al. [23] described a two-stream architecture that initially employs pretrained CNNs, such as Resnet for feature extraction, followed by feature fusion using PCA, convolutional layers, and average pooling [24]. Yu et al. [25] proposed BiMobileNet, an efficient bilinear model utilizing MobileNet for feature extraction and a bilinear pooling module to overcome the LH-CS problem, especially effective with small training samples.

While these methods have demonstrated strong performance, they often overlook the influence of resizing HRSI to fit pretrained CNNs, which can lead to significant information loss. Addressing this, Xie et al. [26] introduced a scale-free CNN (SF-CNN) model, where FCLs are converted into convolutional layers to handle scale-invariant inputs, showing that maintaining the original image size can offer critical information. This underscores the need for further exploration of CNN-based models that can effectively extract discriminative features directly from original-size HRSI, without resizing.

B. Attention Mechanism

Attention mechanisms, which prioritize salient information while filtering out the irrelevant, have shown significant utility in various computer vision tasks. Lin et al. [27] developed a U-Net-based model known as RefineU-Net, which incorporates a residual attention gate module for local refinement in image segmentation. To address class imbalance in image segmentation, Gao et al. [28] developed a multiscale fused model

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termed MSF-ACSA, which employs additive channel–spatial attention to enhance the feature-learning capacity of the model. Building on the success of attention mechanisms in computer vision, their application has extended to HRSI analysis. Wang et al. [29] devised end-to-end recurrent CNNs based on attention mechanism for HRSI-SC, which adaptively selects attention regions. In addition, an attention-consistent network with a dual-branch structure, ACNet, was proposed in [30], utilizing an attention mechanism to extract salient regions and decipher semantic information from HRSI. Thus, the attention mechanism is a useful tool to learn important and effective information within a large image domain.

C. Manifold Learning

Manifold learning, another effective tool for image analysis, aims to discover the low-dimensional structures embedded within high-dimensional datasets, based on the assumption that data points reside on a low-dimensional manifold within a higher dimensional space [31]. Several representative algorithms have been proposed, including the graph-based manifold learning algorithm and Laplacian eigenmaps [32], which have been widely used in feature extraction. For example, Wang et al. [33] constructed the manifold learning constraints both in feature space and label space based on the adaptive graph learning to enhance the feature-learning ability of the model; Hu et al. [34] utilized the adaptive graph embedding to process the whole data to obtain separably low-dimensional image compact features. Graph embedding-based manifold learning aims to preserve the proximity of data points that are close neighbors in the low-dimensional manifold in a local sense. Specifically, assuming that data points $\mathbf{X}_i$ and $\mathbf{X}_j$ are close in the original data distribution, their feature representations in the low-dimensional $\mathbf{Z}_i$ and $\mathbf{Z}_j$ should also be close to each other. The corresponding cost function is defined as follows:

$$\Lambda = {}^1/_2 \sum_{i,j} \mathbf{V}_{ij} \|\mathbf{Z}_i - \mathbf{Z}_j\|_2^2 = \mathrm{tr}\left(\mathbf{Z}^{\mathrm{T}} \mathbf{L}^* \mathbf{Z}\right) \tag{1}$$

where $\mathbf{V}_{ij}$ and $\mathbf{L}^*$ denote the weight matrix and the Laplacian matrix, respectively. Thus, the manifold constraint can be used to ensure that similar images have a close data structure and, in this way, alleviate the LH-CS problem.

III. Proposed Method

In this section, we introduce the motivation for this study and provide a detailed description of the proposed GLR-CNN.

A. Motivation

HRSI scenarios are diverse and complex, often resulting in the LH-CS problem, which poses significant challenges to classification performance. While CNN models are commonly used for HRSI classification, their effectiveness is limited as traditional CNNs cannot process images in their original size. Therefore, how to extract discriminative features from the original size of HRSI is the key issue of classification. Addressing this, we propose the GLR-CNN, a CNN model with global latent relationship embedding, as more detailed in Fig. 1. This model comprises three key components: a deep feature extraction network based on the VGG16 model for learning deep representative features from arbitrary size images, a multilayer feature fusion block with a channel–spatial attention mechanism to obtain complementary image features by integrating crucial information from different layers, and the FCLs with global latent relationship embedding to enhance the consistency between feature and latent relationship to mitigate the LH-CS problem. To improve the classification performance, the objective function combines latent relationship constraint with cross-entropy loss based on label smoothing.

B. Feature Learning

The architecture of the feature-learning model involves three main parts: the deep CNN-based feature extraction branch, the MFF-CNN block, and the FCLs with global latent relationship embedding.

1) Deep CNN-Based Feature Extraction Branch: Given the insufficient training samples for HRSI and for accelerating the convergence of the GLR-CNN model, we utilize pretrained CNNs as the foundational feature extractors. Over recent decades, various deep CNN-based models such as AlexNet, VGGNets, and ResNets have been developed to extract representative features. Among these models, VGG-16 is selected as the foundational model for feature extraction in this article.

VGG16 has obtained excellent performance in image classification fields, outperforming many conventional CNN models. Consequently, it is frequently chosen as the backbone of many models for effective feature learning. The basic structure of VGG16 includes 13 convolutional layers with 3×3 kernels, five max-pooling layers with 2×2 domain, and 3 FCLs. In our model, VGG16 serves as a feature extractor, pulling deep features from the last several convolutional layers. Convolution can be expressed by the following formula:

$$\mathbf{h}^t = \mathrm{ReLU}\left(\mathbf{W}^t \otimes \mathbf{h}^{t-1} + \mathbf{b}^t\right) \tag{2}$$

where $\mathbf{h}^{t-1}$ and $\mathbf{h}^t$ denote the input features and output features of layer t, and $\mathbf{h}^0$ is the input of first layer. $\mathbf{W}^t$ and $\mathbf{b}^t$ are the convolution kernels and biases for layer t, respectively. $\otimes$ represents CNN convolution, and ReLU is the nonlinear function.

Since VGG16 typically requires fixed-size images, resizing HRSI to fit these dimensions may result in the loss of key information. Thus, our model allows input images of any arbitrary size to retain effective information from the HRSI.

2) Multilayer Feature Fusion Block: To optimize the use of HRSI information and harness the unique features of different layers, a MFF-CNN block is developed, as shown in Fig 2. Initially, a feature-learning module with an attention module CBAM [35] (FC-CBAM) is designed. This module includes a convolutional layer, an attention layer, and an adaptive global average pooling (GAP) layer. The multilevel features are then merged, with shallower layers typically reflecting low-level visual information such as edges, while deeper layers represent more abstract image features. Fusing these features helps convey complementary information, enhancing classification performance.

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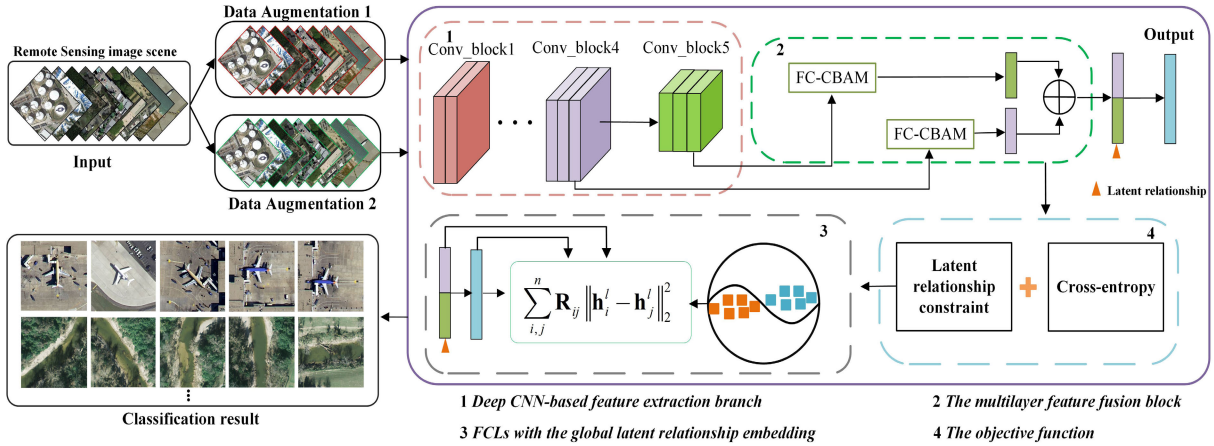


Fig. 1. Workflow of the GLR-CNN framework for HRSI scene classification.

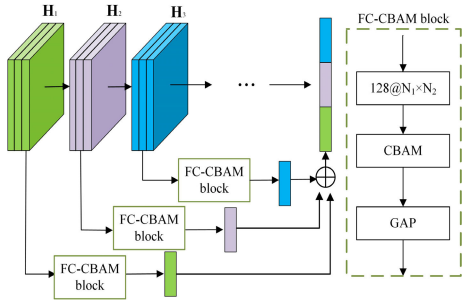


Fig. 2. Architecture of the MFF-CNN block.

The FC-CBAM extracts multilayer features from VGG16, as detailed in Table I. $\mathbf{h}^t$ represents the convolutional layer feature map from VGG16; $\mathbf{W}'$ and $\mathbf{b}'$ represent the convolution kernels and biases of the convolutional layer in FC-CBAM, respectively. $\mathbf{H}'$ denotes the output of the convolutional layer in FC-CBAM, serving as the input to the attention layer, and undergoes both channel and spatial attention, obtaining the key feature map $\mathbf{H}''_{\mathrm{S}}$. Finally, enter $\mathbf{H}''_{\mathrm{S}}$ into the GAP layer to acquire the feature vector. As illustrated in Table I, the attention mechanism in the CBAM is designed to enhance focus on semantically relevant regions, both channelwise and spatially. The CBAM operates as follows:

$$\begin{aligned}\mathbf{H}''_{\mathrm{C}} &= \mathrm{M_C}\left(\mathbf{H}'\right) \odot \mathbf{H}' \\ \mathbf{H}''_{\mathrm{S}} &= \mathrm{M_S}\left(\mathbf{H}''_{\mathrm{C}}\right) \odot \mathbf{H}''_{\mathrm{C}}\end{aligned} \tag{3}$$

where $\mathbf{H}' \in \mathbb{R}^{\mathrm{C}\times\mathrm{O}\times\mathrm{P}}$ denotes the input feature map, $\odot$ represents the elementwise product, $\mathbf{H}''_{\mathrm{C}}$ denotes the output after channel attention, and $\mathbf{H}''_{\mathrm{S}}$ is the final output feature map after applying both channel and spatial attention.

$\mathrm{M_C} \in \mathbb{R}^{\mathrm{C}\times1\times1}$ denotes the channel attention weight coefficient, derived from a shared network consisting of multiplayer perceptron (MLP). This network is defined as follows:

$$\mathrm{M_C}\left(\mathbf{H}'\right) = \sigma\left(\mathrm{MLP}\left(\mathrm{AvgPool}\left(\mathbf{H}'\right)\right) + \mathrm{MLP}\left(\mathrm{MaxPool}\left(\mathbf{H}'\right)\right)\right) \tag{4}$$

where σ denotes the nonlinear activation function, and AvgPool and MaxPool represent the average and maximum pooling features on the input feature, respectively.

$\mathrm{M_S} \in \mathbb{R}^{1\times\mathrm{O}\times\mathrm{P}}$ denotes the spatial attention adjusted by weight coefficient, obtained via a convolution layer employing a 7×7 kernel and computed by

$$\mathrm{M_S}\left(\mathbf{H}''_{\mathrm{C}}\right) = \sigma\left(\mathrm{f}^{7\times7}\left[\mathrm{AvgPool}\left(\mathbf{H}''_{\mathrm{C}}\right); \mathrm{MaxPool}\left(\mathbf{H}''_{\mathrm{C}}\right)\right]\right). \tag{5}$$

The model utilizes a skip connection strategy to fuse multilayer features effectively, where the aggregation of these features is expressed as follows:

$$\mathbf{H}_{\mathrm{con}} = \mathrm{Concat}[\mathbf{h}_1; \mathbf{h}_2; \ldots; \mathbf{h}_{\mathrm{i}}; \ldots; \mathbf{h}_{\mathrm{n}}] \in \mathbb{R}^{1\times1\times3\mathrm{D}} \tag{6}$$

where $\mathbf{h}_i$ denotes the discriminative features from different layers extracted through FC-CBAM.

3) FCLs With Global Latent Relationship Embedding: The latent relationship is utilized to describe the relationship between the latent features of the original images and is embedded within the FCLs to overcome the challenges of LH-CS. FCLs with global latent relationship embedding are integral to enhancing the feature-learning ability of the proposed model.

Typically, FCLs in CNN models are formulated as follows:

$$\mathbf{h}^{l+1} = \mathrm{ReLU}\left(\mathbf{W}^{l+1}\mathbf{h}^{l} + \mathbf{b}^{l+1}\right) \tag{7}$$

where $\mathbf{W}^{l+1}$ and $\mathbf{b}^{l+1}$ denote the weight matrix and biases, respectively. $\mathbf{h}^l$ and $\mathbf{h}^{l+1}$ denotes the nonlinear feature representation at l and $l+1$ layer, respectively.

To improve the consistency between the feature and latent relationship, we construct the global latent relationship and embedded into the FCLs; therefore, how to accurately obtain the latent relationship is a key issue. In this article, we adopt a kind of similarity constructing method based on the introduced global distance distribution [36]. First, calculate the distance between the target image and the others in the dataset, which requires large computer storage. Inspired by recent methodologies [37], data storage is optimized using a queue system constructed through a number of batch data and shared the network between the input batch data and the queue; in this way, distance can be calculated in real time. Then, we need

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TABLE I
ARCHITECTURE OF FC-CBAM

Layer name	Operation		Mathematical formula
Convolutional layer	Convolution ($128@N_1 \times N_2$)		$\mathbf{H}' = \mathrm{ReLU}(\mathbf{W}' \otimes \mathbf{h}^l + \mathbf{b}')$
Attention layer	Channel attention	Multilayer Perceptron	$\mathbf{H}''_C = \mathrm{M_C}(\mathbf{H}') \odot \mathbf{H}'$
	Spatial attention	Convolution (7×7)	$\mathbf{H}''_S = \mathrm{M_S}(\mathbf{H}''_C) \odot \mathbf{H}''_C$
GAP layer	Global AvgPool(p=N)	AvgPool (p=N)	$\mathbf{h}_k = \frac{1}{\lvert N \rvert} \sum_{(p,q)\in N} \mathbf{H}''_{kpq}$

to calculate the distance using the global distance distribution. According to [36], to simplify calculations, a statistical distance histogram replaces the detailed global distance distribution, $\mathrm{P}_i(d) = (\mathrm{sum}(\mathrm{S}_i, (d, d+1))/N$ represents the distance probability mass function at the dth interval, and sum() counts the number of points in S_i whose distances fall into the range of $(d,\ d+1)$. $\mathrm{D_{KL}}(\mathbf{h}_i^l || \mathbf{h}_j^l)$ represents the Kullback–Leibler (KL) divergence [38], used to measure the difference between two different distance histograms, and is calculated as follows:

$$\mathrm{D_{KL}}\left(\mathbf{h}_i^l || \mathbf{h}_j^l\right) = \sum_{d=1}^{D} \mathrm{P}_i(d) \log \frac{\mathrm{P}_i(d)}{\mathrm{P}_j(d)}. \tag{8}$$

Subsequently, the KL divergence is used to optimize the traditional distance between $\mathbf{h}_i^l$ and $\mathbf{h}_j^l$. $\mathrm{D}(\mathbf{h}_i^l, \mathbf{h}_j^l)$ represents the distance based on the global distance distribution between $\mathbf{h}_i^l$ and $\mathbf{h}_j^l$ and can be computed by

$$\mathrm{D}\left(\mathbf{h}_i^l, \mathbf{h}_j^l\right) = \mathrm{Dis}\left(\mathbf{h}_i^l, \mathbf{h}_j^l\right) \times \left(2 - \mathrm{D_{KL}}\left(\mathbf{h}_i^l || \mathbf{h}_j^l\right)\right) \tag{9}$$

where $\mathrm{Dis}(\mathbf{h}_i^l, \mathbf{h}_j^l)$ denotes the cosine distance between $\mathbf{h}_i^l$ and $\mathbf{h}_j^l$.

Therefore, the latent relationship can be computed by

$$\mathbf{R}_{ij} = \exp\left(-\mathrm{D}\left(\mathbf{h}_i^l, \mathbf{h}_j^l\right)\right) \tag{10}$$

where $\mathbf{R}_{ij}$ denotes the latent relationship between the ith and jth features in the lth layer.

A Laplacian Eigenmap algorithm is employed to model the relationship between the latent relationship and image features. It is constructed by a graph, where each node represents the latent features, and each edge represents the latent relationships between them. This setup presumes that if two images exhibit a high relationship, they should share a similar structure in their low-dimensional latent features. The constraint is, thus, formulated as follows:

$$\Theta_{\mathrm{LR}} = \frac{1}{2} \sum_{l=1}^{L} \sum_{i,j}^{n} \mathbf{R}_{ij}^l \| \mathbf{h}_i^{l+1} - \mathbf{h}_j^{l+1} \|_2^2 \tag{11}$$

where $\mathbf{R}$ denotes the latent relationship, n represents the number of samples, and L denotes the total number of the FCLs.

C. Loss Function

To enhance the generalization of the GLR-CNN model, label smoothing is adopted to boost the efficacy of the cross-entropy loss. Typically, for HRSI-SC, the CNN model undergoes fine-tuning using a limited number of labeled data samples $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_i, \ldots, \mathbf{x}_n\}$, and the number of training samples is denoted by n. During this fine-tuning phase, the cross-entropy loss, augmented by label smoothing, is used to train the classifier to accurately categorize the HRSI scenes in the article. The loss function is defined as

$$\Theta_C = -\frac{1}{n} \sum_{i=1}^{n} \sum_{k=1}^{K} y_{ik} \log(f(x_{ik})) \tag{12}$$

where $f(x_k)$ is the output of the model, denoting the probability assigned to the kth class, K is the number of categories, y_{ik} is "$1-\kappa$" for the correct class, and "$\kappa/(K-1)$" for the rest, κ is a small constant.

Generally, the objective function of the GLR-CNN model can be formulated as follows:

$$\begin{aligned} \Theta &= \Theta_{\mathrm{C}} + \lambda \Theta_{\mathrm{LR}} \\ &= -\frac{1}{n} \sum_{i=1}^{n} \sum_{k=1}^{K} y_{ik} \log(f_{ik}) + \frac{\lambda}{2} \sum_{l=1}^{L} \sum_{i,j=1}^{n} \mathbf{R}_{ij}^l \| \mathbf{h}_i^{l+1} - \mathbf{h}_j^{l+1} \|_2^2. \end{aligned} \tag{13}$$

The procedure of HRSI-SC using the GLR-CNN model is summarized in Algorithm 1.

Algorithm 1 GLR-CNN Model for HRSI-SC

Input: Input HRSI scene; number of iterations T; learning rate α; number of image of the queue S; weight parameter of the latent relationship constraint λ.
Output:Predict label for input HRSI scene.
Algorithm:

1: **Initialization**: The parameters of pretrained VGG16 model's convolution layer, i.e., $\mathbf{W}^t$, $\mathbf{b}^t$; the convolution layer parameters ($\mathbf{W}'$ and $\mathbf{b}'$) in the MFF-CNN, and the parameters in the CBAM; $\mathbf{W}^l$ and $\mathbf{b}^l$ of the FCLs, and iter=0.
2: **While** iter $\leq$ iterations :
3: Calculate the loss according to formula (13);
4: Update $\mathbf{W}'$, $\mathbf{b}'$, $\mathbf{W}^l$, $\mathbf{b}^l$ and the parameters in the CBAM by back-propagation using the stochastic gradient descent algorithm;
5: Update iter=iter+1.
6: **End while**

IV. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, to evaluate the validity of the proposed model, it is compared with different state-of-the-art methods on HRSI datasets: UC Merced Land-Use Dataset (UCM) [39], aerial image dataset (AID) [40], and NWPU-RESISC45 (NWPU) [6]. The results demonstrate that the GLR-CNN method can surpass most of these methods and prove its effectiveness.

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Fig. 3. Sample images from the UCM dataset. From top left to bottom right: agricultural area, airplane, baseball diamond, beach, buildings, chaparral, dense residential area, forest, freeway, golf course, harbor, intersection, medium density residential area, mobile home park, overpass, parking lot, river, runway, sparse residential area, storage tanks, and tennis courts.

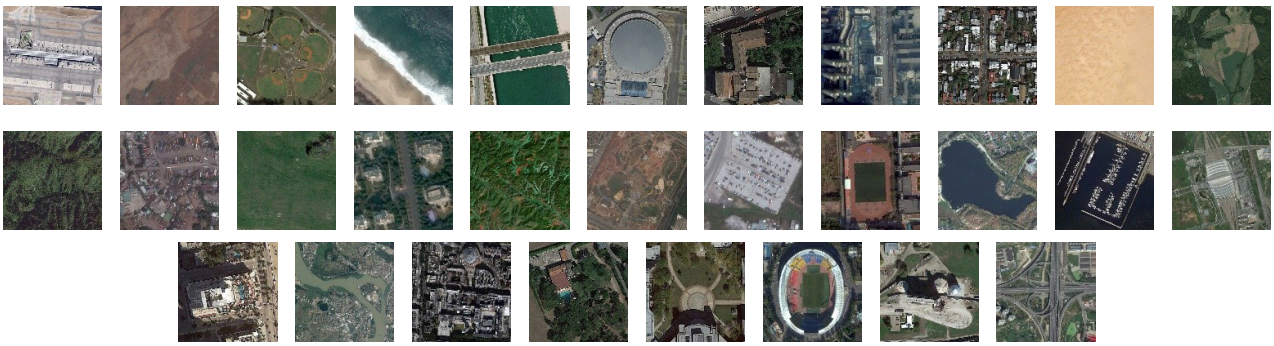


Fig. 4. Sample images from the AID dataset. From top left to bottom right: airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, and viaduct.

A. Datasets

UCM: The UCM dataset comprises 2100 images, each with a fixed size of 256 × 256 pixels. It features 21 typical land-use categories, each containing 100 images with a resolution of 30 cm. Sample images of each category are illustrated in Fig. 3.

AID: The AID contains 10 000 images, each measuring 600 × 600 pixels across 30 scene categories. Each land-use class includes hundreds of images (ranging from 220 to 420) with multiple resolutions from 8 to 0.5 m. Sample images of each category are illustrated in Fig. 4.

NWPU: The NWPU includes 31 500 images, each 256 × 256 in size, divided into 45 categories. Each land-use category contains 700 images with a resolution ranging from approximately 0.3 to 0.2 m. Sample images of each category are illustrated in Fig. 5.

B. Evaluation Metrics

For HRSI-SC, overall accuracy (OA) and confusion matrix (CM) are the common evaluation metrics. OA measures the correct prediction rate of the model for the training samples. CM is employed to intuitively show the performance of the model by category, where each row represents the predicted label for each category and each column represents the corresponding true label.

C. Experimental Setup

In this article, all experiments with the proposed method are conducted using the PyTorch framework, with the following computer configuration: Intel Xeon E5-2640 CPU and NVIDIA RTX 2080Ti GPU. The network hyperparameter settings include the following: momentum coefficient is 0.9 and weight decay is 0.0005; the initial learning rate is set at 0.03, which is reduced by a factor of ten at epochs 60 and 80; the batch size is 32; and the maximum number of epochs is set to 100. Training samples are augmented using random color distortions, random Gaussian blur, and horizontal flipping. All experimental results represent the average of ten experiments.

To test the effectiveness of the GLR-CNN model, the dataset is split using two different training ratios for comparison. For the UCM dataset, training ratios are set at 50% and 80%; for the AID dataset, at 20% and 50%; and for the NWPU dataset, at 10% and 20%, with the remaining samples used for testing.

D. Comparisons With the Other CNN-Based Methods

We conducted numerous comparative experiments against several CNN-based methods across three different datasets to verify the effectiveness of the GLR-CNN model. MFF-CNN represents VGG16 integrated with a multilayer feature fusion block, and fine-tuned VGG16 represents a network without a multilayer feature fusion block and latent relationship constraints.

1) UCM Dataset: In this experiment, the GLR-CNN was compared against seven mainstream methods to verify its effectiveness, with results summarized in Table II. Compared with the fine-tuned VGG16 [40], the OA of our method has increased by 3.38% and 3.93%, respectively, demonstrating significant improvements with the GLR-CNN model. Furthermore, the performance of MFF-CNN improved by 2.6% and 3.12% compared to the ordinary Fine-tuned VGG16; compared with MFF-CNN, the performance of GLR-CNN improved by 0.78% and 0.81%, respectively. These

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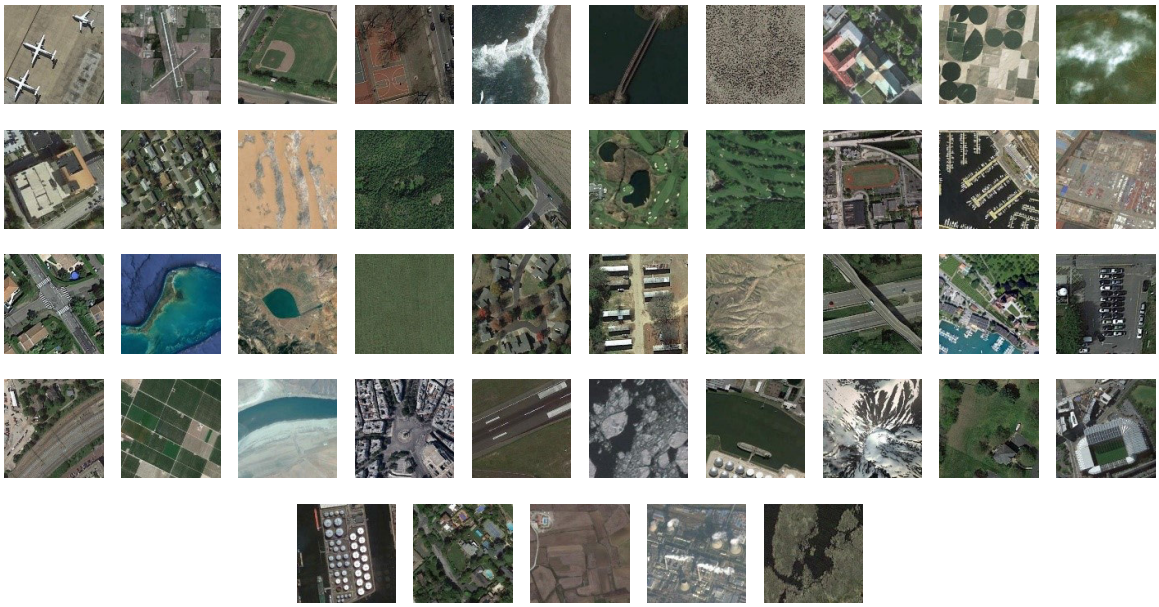


Fig. 5. Sample images from the NWPU dataset. From top left to bottom right: airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snow berg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland.

TABLE II

OA (%) OF DIFFERENT METHODS UNDER TRAINING RATIOS OF 50% AND 80% ON THE UCM DATASET

Basic Network	Methods	Training Ratio	
		50%	80%
VGG16	GLR-CNN(ours)	**97.52± 0.33**	**99.14± 0.26**
VGG16	MFF-CNN(ours))	96.74	98.33
VGG16	Fine-tuned VGG16 [40]	94.14±0.69	95.21±1.20
VGG16	SF-CNN [26]	/	99.05±0.27
CaffeNet, VGG16, and GoogLeNet	Two-stream fusion [41]	96.97±0.75	98.02±1.03
VGG16	ARCNet [29])	96.81±0.14	99.12±0.40
VGG16	DCNN [21])	/	98.93±0.10
VGG16	VGG16-CapsNet [22]	95.33±0.18	98.81±0.22
VGG16	MSCP [42]	/	98.36±0.58
VGG16	MGDNet [43]	97.48±0.32	/

The best result are highlight in bold.

results underscore the impact of the latent relationship constraint and the MFF-CNN in enhancing scene classification performance.

As depicted in Figs. 6 and 7, which, respectively, show the CM generated by the GLR-CNN with 50% and 80% training ratios, it can be observed from Fig. 6 that the GLR-CNN model successfully identifies most scene categories except for the dense residential and medium residential. Here, 10% of the medium residential is misclassified as dense residential, and 12% of dense residential is misclassified as medium residential due to the large interclass similarities in building distribution. With an increase in training samples, as shown in Fig. 7, this problem is significantly improved. This analysis shows that the GLR-CNN model is effective for scene classification in the UCM dataset.

2) AID Dataset: Table III shows a comparative evaluation of seven CNN-based methods for HRSI-SC on the AID dataset. The OA obtained by our proposed method reached 95.64% and 93.09% with 50% and 20% training rates, respectively. Although not the optimal performance, it surpasses most other methods. Compared with the Fine-tuned VGG16 model [40], performance improved by 6.5% (20%) and 6% (50%), respectively. The ability of our model to extract more effective features and achieve superior scene classification is further demonstrated as the performance of MFF-CNN surpasses the ordinary Fine-tuned VGG16, and GLR-CNN outperforms MFF-CNN, showing that latent relationship constraints and the multilayer feature fusion block can effectively improve the capabilities of deep CNN model for HRSI-SC.

From Figs. 8 and 9, it can be observed that the CM was generated by GLR-CNN with 20% and 50% training ratios, respectively. Fig. 8 shows considerable confusion among the center, school, and square due to large intraclass diversity. Furthermore, parks and resorts are also prone to misclassification due to their significant interclass similarity. Fig. 9 compares these results with other methods, showing that the OA for center (76%), school (77%), and square (83%) in ARCNet [29] is lower than that for the OA of center (83%), school (87%), and square (85%) in the GLR-CNN model for these categories. The proposed method GLR-CNN effectively captures image

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TABLE III
OA (%) OF DIFFERENT METHODS UNDER TRAINING RATIOS OF 20% AND 50% ON THE AID DATASET

Basic Network	Methods	Training Ratio	
		20%	50%
VGG16	GLR-CNN(ours)	93.09±0.16	95.64±0.11
VGG16	MFF-CNN(ours))	91.23	94.14
VGG16	Fine-tuned VGG16 [40]	86.59±0.29	89.64±0.36
VGG16	SF-CNN [26]	**93.12±0.12**	96.66±0.11
CaffeNet, VGG16, and GoogLeNet	Two-stream fusion [41]	92.32±0.41	94.58±0.25
VGG16	ARCNet [29])	88.75±0.40	93.10±0.55
VGG16	DCNN [21])	90.82±0.16	**96.89±0.10**
VGG16	VGG16-CapsNet [22]	91.63±0.19	94.74±0.17
VGG16	MSCP [42]	91.52±0.21	94.42±0.17
VGG16	MGDNet [43]	86.52±0.81	92.41±0.57

The best result are highlight in bold.

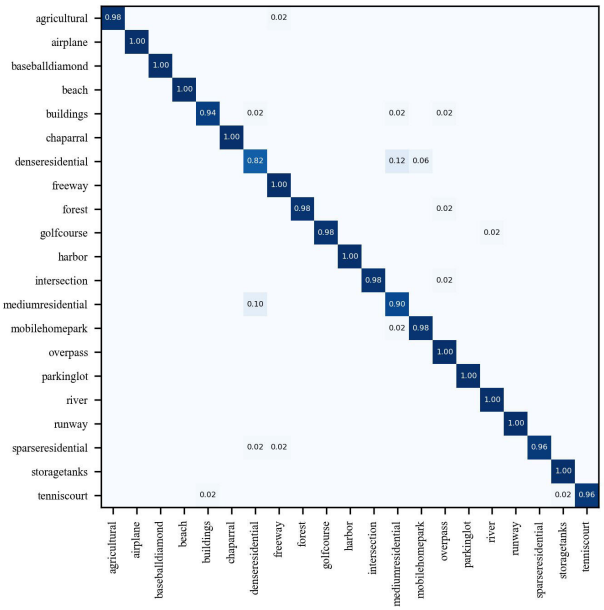


Fig. 6. CM on the UCM dataset with a training ratio of 50%.

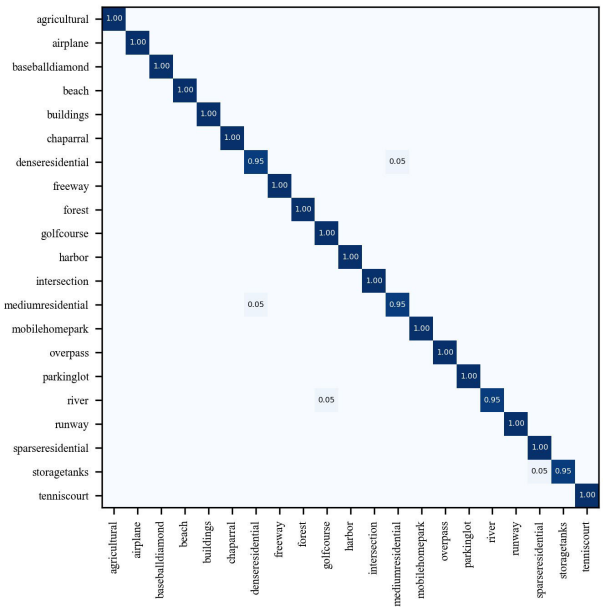


Fig. 7. CM on the UCM dataset with a training ratio of 80%.

correlations and enhances the ability to distinguish categories with significant intraclass diversity and interclass similarity.

3) NWPU Dataset: This challenging dataset also underwent evaluation. As shown in Table IV, we compared our method against six mainstream CNN-based scene classification methods. The OAs achieved were 89.35% and 92.11% with 10% and 20% training ratios, respectively. Although not the best and lower than SF-CNN [26], our performance exceeds most other methods. Notably, compared with Fine-tuned VGG16, GLR-CNN improved OA by 2.2% at a 10% training rate and 1.75% at a 20% rate. While MFF-CNN did not show improvement over Fine-tuned VGG16 due to the extensive LH-CS problems in the NWPU dataset, GLR-CNN performed better, validating that embedding a model of latent relationships can significantly enhance classification efficacy. These results affirm that the GLR-CNN model is robust for scene classification tasks across diverse and complex datasets.

Figs. 10 and 11 display the CM generated by the GLR-CNN method with 10% and 20% training ratios. As observed from Fig. 10, certain category groups are prone to confusion due to high interclass similarities in the NWPU dataset, such as churches, commercial areas, palaces, railways, and railway stations; rectangular farmlands and terraces; and rivers, lakes, and wetlands. As shown in Fig. 11, notably, the performance of these categories has improved; churches and palaces exhibit the lowest OA with 76% and 75%, respectively, yet these scores surpass most comparative methods. From the results, we observe that the GLR-CNN model effectively addresses the challenges of large-scale NWPU datasets by reducing visual confusion and enhancing feature recognition capabilities.

E. Parameter Analysis

In the GLR-CNN model, three parameters—the size of the input image, the number of layers in MFF-CNN, and the weight parameter of the latent relationship constraint—are tested on three datasets with different training ratios to analyze their influence on classification performance. In subsequent experiments, all other parameters are set to the optimal values to focus on evaluating these target parameters.

1) Image Size of Input Image: To determine whether input image size significantly impacts scene classification

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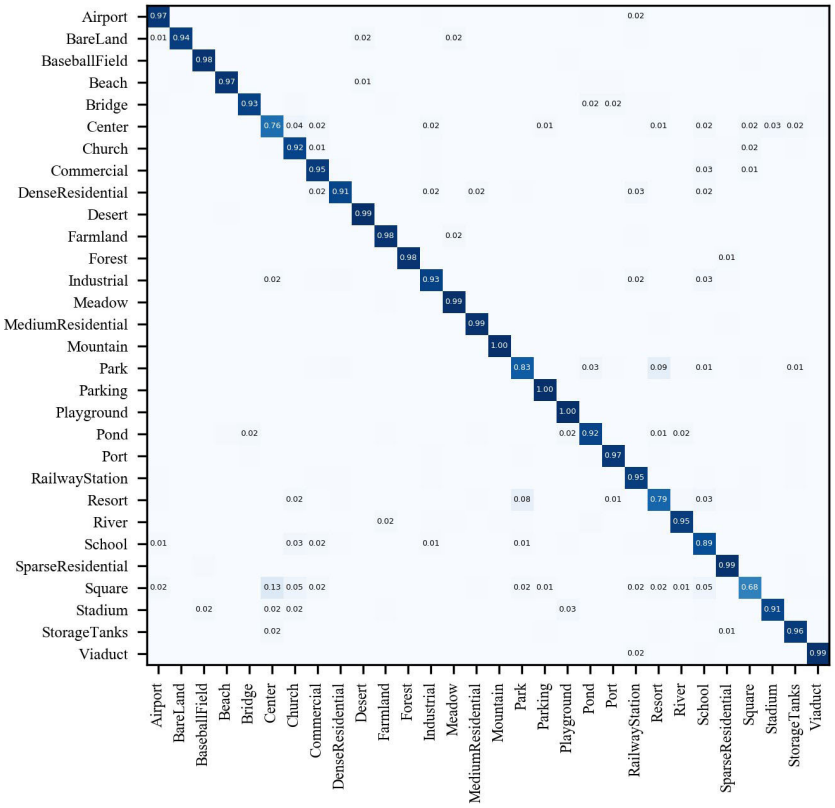


Fig. 8. CM on the AID dataset with a training ratio of 20%.

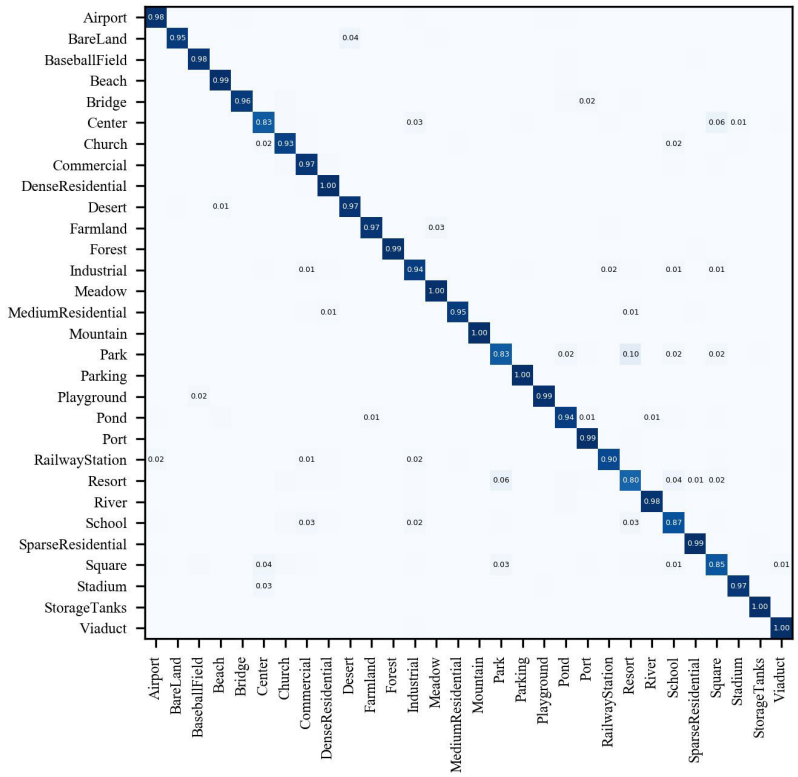


Fig. 9. CM on the AID dataset with a training ratio of 50%.

accuracy, input images of various sizes were tested. Two sizes, 224×224 and 256×256, were used for the UCM and NWPU datasets, and four sizes, 224×224, 256×256, 300×300, and 400×400, were used for the AID dataset. Results from Table V

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TABLE IV

OA (%) OF DIFFERENT METHODS UNDER TRAINING RATIOS OF 10% AND 20% ON THE NWPU DATASET

Basic Network	Methods	Training Ratio	
		10%	20%
VGG16	GLR-CNN(ours)	89.35± 0.25	92.11±0.22
VGG16	MFF-CNN(ours))	86.07	89.88
VGG16	Fine-tuned VGG16 [6]	87.15±0.45	90.36±0.18
VGG16	SF-CNN [26]	**89.89±0.16**	**92.55±0.14**
CaffeNet, VGG16, and GoogLeNet	Two-stream fusion [41]	80.22±0.22	83.16±0.18
VGG16	DCNN [21])	89.22±0.50	91.89±0.22
VGG16	VGG16-CapsNet [22]	85.08±0.13	89.18±0.14
VGG16	MSCP [42]	85.33±0.17	88.93±0.14
VGG16	MGDNet [43]	84.81±0.36	91.41±0.69

The best result are highlight in bold.

TABLE V

OA (%) OF IMAGE SIZES ON DIFFERENT DATASETS OF A TABLE

Image size	UCM		AID		NWPU	
	50%	80%	20%	50%	10%	20%
224 × 224	97.14	98.80	90.43	93.86	88.26	90.74
256 × 256	**97.52±0.33**	**99.14±0.26**	92.13	94.14	**89.35±0.25**	**92.11±0.22**
300 × 300	/	/	92.73	94.96	/	/
400 × 400	/	/	**93.09±0.16**	**95.64±0.11**	/	/

The best result are highlight in bold.

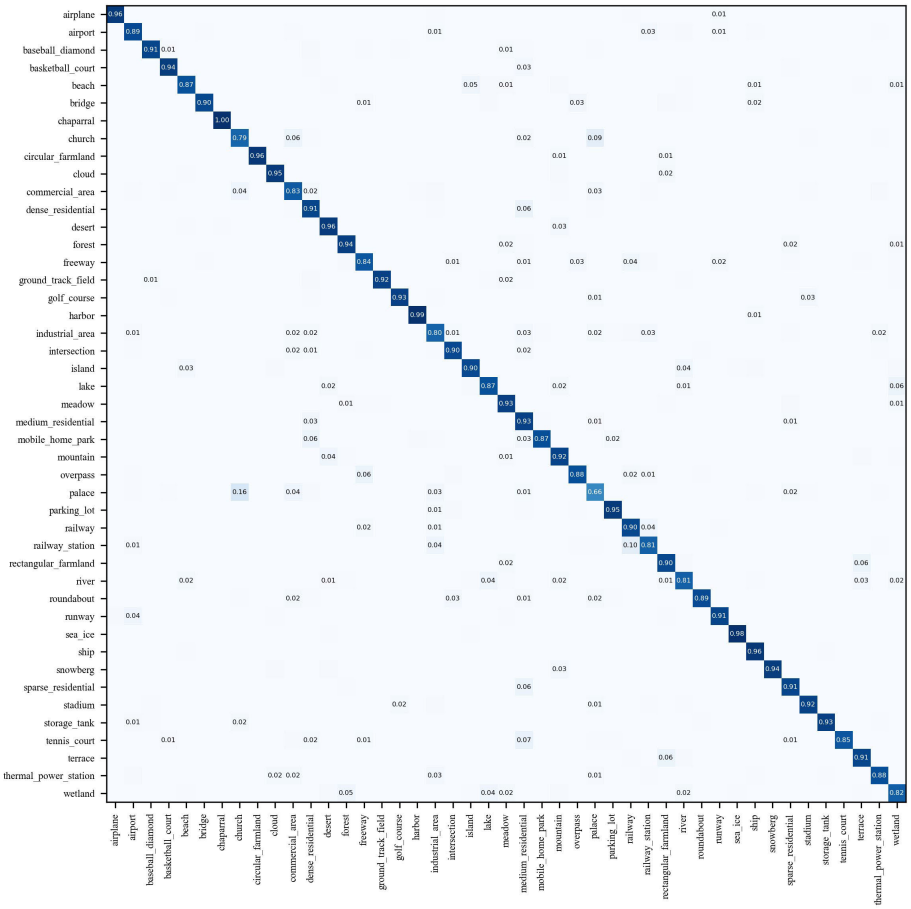


Fig. 10. CM of the NWPU dataset with a training ratio of 10%.

clearly show that the GLR-CNN model performs better when the input size matches or is close to the original size of the HRSI, which indicates that larger input sizes retain more useful feature information.

2) Number of Layers in MFF-CNN: Features from different CNN layers contain complementary information. We tested the multilayer feature-learning capability of the model by comparing the features of cross-level connections between two

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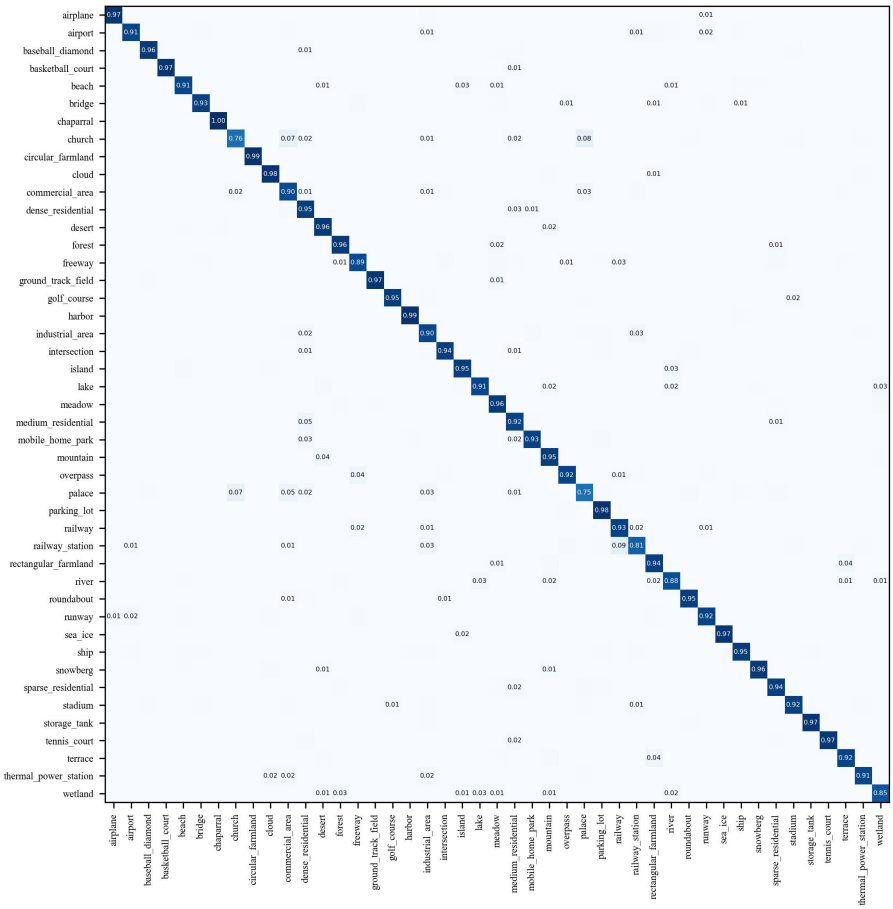


Fig. 11. CM of the NWPU dataset with a training ratio of 20%.

TABLE VI
OA (%) OF FUSING DIFFERENT LAYERS ON DIFFERENT DATASETS

Number of layers in skipping connections	UCM		AID		NWPU	
	50%	80%	20%	50%	10%	20%
2	**97.52±0.33**	**99.14±0.26**	**93.09±0.16**	**95.64±0.11**	**89.35±0.25**	**92.11±0.22**
3	96.76	98.09	91.53	94.8	87.91	91.47

The best result are highlight in bold.

levels of VGG16, conv_block 4-3 and conv_block 5-3, and three levels, conv_block 3-3, conv_block 4-3, and conv_block 5-3. The classification results, as presented in Table VI, reveal that increasing the number of connected layers does not enhance performance. This may be due to the lower level features not adequately capturing critical scene information and instead introducing noise.

3) Weight Parameter of Latent Relationship Constraint: The importance of latent relationship learning in HRSI-SC was evaluated by varying the weight parameter of the latent relationship constraint in the objective function across a range of 1e-1-1, conducting experiments at different training ratios. The experimental results of 80% UCM, 20% AID, and 10% NWPU training ratios are shown in Figs. 12–14, respectively, illustrating that the OA varies significantly with changes in the parameter value, with maximum differences in classification accuracy for GLR-CNN across the datasets being 1.67%, 1.6%, and 2.72%, respectively. This variability indicates that the OA of GLR-CNN is sensitive to parameter settings. Therefore, we selected the parameter value that yielded the best classification results for implementation.

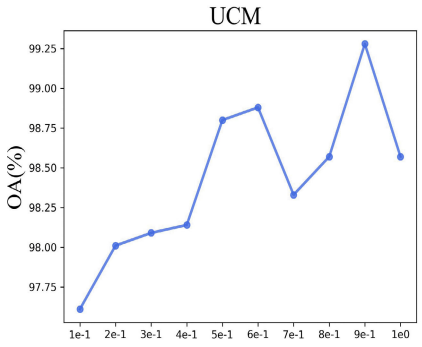


Fig. 12. Influence of parameter λ on OA in the UCM dataset.

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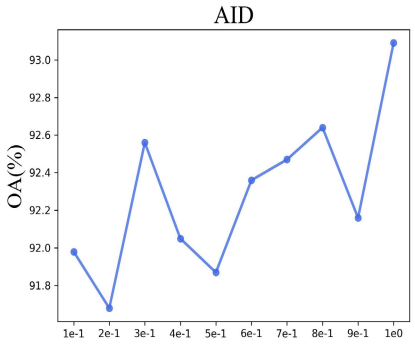


Fig. 13. Influence of parameter λ on OA in the AID dataset.

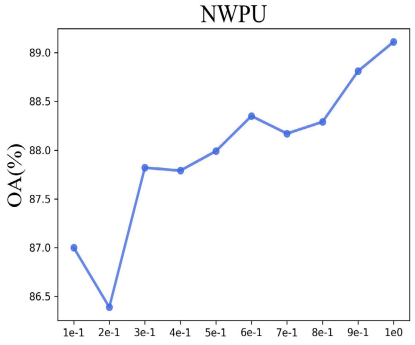


Fig. 14. Influence of parameter λ on OA in the NWPU dataset.

V. Conclusion

In this work, to enhance the feature-learning capacity of deep CNN from the original size image, we proposed a novel CNN-based model for HRSI-SC. Based on the deep VGG16 network, combine an MFF-CNN block with a channel–spatial attention algorithm with the FCLs with the global latent relationship embedding. The MFF-CNN block merges the key information of different levels in the deep VGG16 network, which not only extracts the representative features among different levels but also enhances the model to learn important information in a large-scale feature map. Based on the extracted representative features, FCLs with the global latent relationship embedding construct the latent relationship constraint to enhance the consistency between the image features and latent relationship and alleviate the LH-CS problem further. We evaluated the capability of the GLR-CNN model and each part of the model on three public HRSI datasets and achieved great performance. In future work, we will investigate other deep learning models that can be directly applied to the original size HRSI for high-precision scene classification feasibility; furthermore, hypergraph learning can be used to accurately construct the latent relationship between image features and improve the deep CNN discriminative feature-learning ability from HRSI of the original size.

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